

SKILL SUMMARY

- **Languages:** Python, JavaScript, SQL, C++, Java, HTML, CSS.
- **Libraries:** YOLOv8, OpenCV, MediaPipe, NumPy, Pandas, Scikit-learn.
- **Tools:** Git, GitHub, Google Colab, Visual Studio Code.
- **Cloud:** AWS (EC2, S3, IAM), Docker (Basics), DigitalOcean.
- **Soft Skills:** Communication, Problem-Solving, Time Management, Teamwork, Adaptability.

WORK EXPERIENCE

AI/ML Project Intern

KalydoAi, Rajkot, Gujarat

[Nov 2025 – January 2026]

- Developed and tested computer vision models for real-world applications
- Implemented image processing pipelines using OpenCV
- Improved model performance through dataset cleaning and optimization
- Collaborated on debugging and deployment workflows

Software Developer Intern

Xpertnest (Remote)

[May 2023 – June 2023]

- Developed and maintained software features as part of a remote development team
- Worked on coding, debugging, and improving application functionality
- Collaborated using version control and followed standard development practices

PROJECTS

Real-Time Hand Motion Interaction System [Repo](#)

Tech Stack: Python, OpenCV, MediaPipe

Using Python, OpenCV, and MediaPipe, I developed a real-time hand gesture recognition system that allows users to control and play games directly through hand movements. The project leveraged MediaPipe's hand-tracking and OpenCV's image processing capabilities to ensure accurate detection and tracking.

Real-Time Ambulance Tracking & Smart Signal System [Repo](#)

Tech Stack: Python, YOLOv8 Nano, OpenCV

Using Python, YOLOv8 Nano, and OpenCV, I developed a real-time ambulance detection and traffic signal prioritization system that identifies ambulances from live traffic feeds and automatically switches the corresponding signal to green. The project involved training and fine-tuning the YOLOv8 model for high-accuracy detection, integrating OpenCV for video frame analysis and visualization, and optimizing model performance through dataset cleaning, augmentation, and efficient inference handling.

RESEARCH PUBLICATIONS

AI-Based Smart Traffic Management System for Ambulance Prioritization (Research Paper)

- Published research on an AI-powered traffic management system using YOLOv8 for real-time ambulance detection and intelligent traffic signal prioritization.
- Presented the project to Rajkot Municipal Corporation (RMC) during the AI-Based District Development presentation, demonstrating its potential application in smart city infrastructure. [View Paper](#) →

Python-Based Gesture Controlled Game Using MediaPipe (Research Paper)

Published research on touchless human-computer interaction using Python, MediaPipe, and OpenCV for real-time gesture recognition and game control. [View Paper](#) →

EDUCATION

DIPLOMA IN COMPUTER ENGINEERING

Atmiya University, Rajkot

B.E IN INFORMATION TECHNOLOGY

VVP Engineering College, Rajkot